

The Adelic Physics Program: Epistemological Foundations and Communications Framework

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Abstract

The QNFO Adelic Physics Program proposes that the physically accessible base field of physics is $\mathbb{Q}$ (the rational numbers), not $\mathbb{R}$ (the real numbers). Ostrowski’s theorem — a proven result in number theory — then demands that all p-adic completions of $\mathbb{Q}$ be physically meaningful, transforming the Adelic Physics Program from mathematical speculation into a physical programme. The core scientific claims of this programme are established across 56 published QNFO papers, spanning adelic quantum error correction, Bruhat-Tits physical geometry, ultrametric quantum computation, Langlands physics, and the Gisin–Del Santo critique of real numbers as hidden variables. This paper provides the epistemological and pedagogical infrastructure that the programme has lacked: a single accessible document that explains the argument from foundations to implications. We construct numbers from distinction primitives via the Laws of Form Number Builder, trace the historical stratigraphy of measurement, analyze domain translation errors that have obscured the programme’s reception, and develop an observer-centered epistemology that situates the adelic framework within the broader philosophical tradition of finite-information physics. The intended audience is the working physicist who finds “adelic QFT” opaque — this paper shows that the programme is better understood as the removal of an unjustified assumption (that $\mathbb{R}$ is the natural base field of physics) rather than the addition of new physics.

Keywords: adelic physics, Ostrowski’s theorem, p-adic numbers, Bruhat-Tits trees, Laws of Form, ontological closure, finite information quantities, Langlands program, quantum foundations, epistemology of physics

1. Introduction

Physics today operates exclusively at the ∞ -place — the Archimedean completion $\mathbb{R}$. Quantum mechanics, quantum field theory, general relativity, and the Standard Model are all formulated over the real numbers (or their complex extension $\mathbb{C}$). This choice is so deeply embedded in the practice of physics that most physicists have never considered it a choice at all. It appears as natural as using base-10 arithmetic — a convention so fundamental it is invisible.

This paper argues that the choice of $\mathbb{R}$ as the base field of physics is an unjustified assumption with measurable consequences. The physically accessible numbers are rational — every measurement

produces a rational number, every computer simulation operates on finite-precision approximations, and every number that has ever appeared in a physics paper (π , e , $\sqrt{2}$, the fine-structure constant α) is computable — meaning it can be approximated to arbitrary precision by a finite rational procedure. If $\mathbb{Q}$ is the physically accessible base field, then Ostrowski’s theorem (1916) (Ostrowski 1916) — which classifies all completions of $\mathbb{Q}$ — becomes a physical organizing principle, not merely a mathematical curiosity.

The QNFO Adelic Physics Program has developed this argument across 56 published papers (Quni-Gudzinis 2026a, 2026e, 2026c, 2026g, 2026i, 2026d, 2026h, 2026b; QNFO Research 2026). The core scientific claims are established. What the programme has lacked — and what this paper provides — is a coherent epistemological and pedagogical framework that makes the argument accessible to the working physicist.

This paper is structured as a progressive disclosure. Section 2 identifies $\mathbb{R}$ as an unjustified assumption. Section 3 presents the number-theoretic foundations: Ostrowski’s theorem and Tate’s thesis. Section 4 develops a constructive account of number systems from distinction primitives — the Laws of Form (LoF) Number Builder — and traces the historical stratigraphy of measurement. Section 5 presents the Gisin–Del Santo programme’s independent convergence on the same conclusion: real numbers are the hidden variables of classical mechanics (Gisin 2019; Del Santo and Gisin 2019). Section 6 presents the physical geometry of Bruhat-Tits trees and their connection to Standard Model mass ratios. Section 7 outlines ultrametric quantum computation. Section 8 connects the programme to Langlands physics via S-duality (Kapustin and Witten 2006). Section 9 addresses the pedagogical challenge — domain translation errors — and Section 10 develops an observer-centered epistemology. Section 11 concludes with the implications of taking $\mathbb{Q}$ seriously as the physical base field.

2. The Unjustified Assumption

2.1 $\mathbb{R}$ Is Not Forced by Physical Necessity

The real numbers were developed in the 19th century to provide a rigorous foundation for calculus. Cauchy, Dedekind, and Weierstrass constructed $\mathbb{R}$ to patch the logical gaps in Newton’s and Leibniz’s infinitesimal calculus — to make the continuum “well-behaved” for analysis. Physics inherited $\mathbb{R}$ from mathematics, not from experiment.

There is no experimental measurement that distinguishes a real number from a rational approximation to it. Every measurement produces a rational number to within finite precision. Every computer simulation of physics — from N-body gravitational simulations to lattice QCD to quantum circuit simulators — operates on finite-precision rational numbers (IEEE 754 floating-point, which is a subset of $\mathbb{Q}$). These simulations reproduce experimental results to within measurement precision [established — computational physics practice].

The burden of proof therefore shifts to the advocate of $\mathbb{R}$: demonstrate a physical prediction that requires infinite-precision real numbers and cannot be reproduced by any finite-precision rational computation. To our knowledge, no such prediction exists [untested — open research question].

2.2 The Computable Reals Argument

Every number that has ever appeared in a physics paper — π , e , $\sqrt{2}$, the fine-structure constant α , particle masses, coupling constants — is computable [established]. A computable real number is one for which there exists a finite Turing-machine procedure (a D/R procedure in the terminology of Autaxys Ontological Closure) that approximates it with a computable modulus of convergence. Computable reals form a countable subset $\mathbb{R}_{\text{comp}} \subset \mathbb{R}$.

The non-computable reals (Chaitin's Ω , etc.) have never appeared in any physical prediction [established]. If a physical theory required a non-computable real number, that theory would be fundamentally untestable — no finite measurement could distinguish the predicted value from a computable approximation. A theory that makes untestable predictions fails the falsifiability requirement.

We are therefore justified in restricting physical theories to computable ($\mathbb{Q}$ -approximable) quantities. This is not a limitation — it is a recognition that untestable predictions are not physics [established principle — falsifiability].

2.3 The Map Is Not the Territory

A common objection: “Quantum mechanics is formulated over $\mathbb{C}$, which contains $\mathbb{R}$.” The response is that the *formalism* of quantum mechanics is over $\mathbb{C}$, but the *physical content* is in measurement outcomes, which are rational. The use of complex numbers in the formalism no more requires the physical reality of $\mathbb{R}$ than the use of negative numbers in accounting requires the physical reality of anti-dollars. The map is not the territory.

3. Number-Theoretic Foundations

3.1 Ostrowski's Theorem (1916)

Statement: Every non-trivial absolute value on $\mathbb{Q}$ is equivalent to either: - The standard Archimedean absolute value $|\cdot|_{\infty}$ (whose completion is $\mathbb{R}$), or - A p -adic absolute value $|\cdot|_p$ for some prime p (whose completion is $\mathbb{Q}_p$).

Physical significance: There is no “continuous scale” axiom built into nature. $\mathbb{R}$ is a choice of completion, not a necessity. This theorem is proven — it is not a conjecture or an interpretation [established theorem]. It is the anchor of the entire consilience framework.

If $\mathbb{Q}$ is the physical base field, then Ostrowski's theorem forces the following conclusion: the Archimedean completion $\mathbb{R}$ and every p -adic completion $\mathbb{Q}_p$ are equally legitimate completions of the physical number system. Physics has arbitrarily restricted itself to the ∞ -place.

3.2 Tate's Thesis as a Template

Tate's 1950 thesis (Tate 1950) reformulated Hecke L-functions (including the Riemann zeta function) using adelic Fourier analysis. The key insight: the functional equation $\zeta(s) = \zeta(1-s)$ of the Riemann zeta function emerges naturally as a consequence of adelic Poisson summation. Local zeta integrals at each place (∞ , 2, 3, 5, ...) multiply to give a global zeta integral over the idele class group, from which the functional equation follows as a single adelic identity.

Physical significance: Tate’s thesis provides the template for a “global” formulation of any field theory. If the dynamics of a quantum system factorize into local contributions at each completion of $\mathbb{Q}$, then global consistency — the functional equation — emerges from the adelic product structure. The Riemann zeta function’s analytic continuation is the statement that spectral data at all finite primes must be consistent with the Archimedean data at $s = 1$.

The adelic product formula — $\prod_v |x|_v = 1$ for all $x \in \mathbb{Q}^\times$ — is the number-theoretic analogue of a conservation law. Every rational number has a total “valuation” of 1 across all places simultaneously.

4. Constructing Numbers from Distinctions

4.1 The Laws of Form Number Builder

Numbers are not Platonic givens. They are constructed from the two primitives of distinction — marking and enclosing — through a finite sequence of operations. The LoF Number Builder (Quni-Gudzinis 2026) formalizes this construction in six steps:

Step	Operation	Result
1. Draw a distinction	Apply $\#$ once	The number 1
2. Calling	$\#\# = \#$, iterated	$\mathbb{N}$
3. Silent Radix	Positional notation from nested enclosures	Decimal representation
4. Enclose groups	Nested $[]$ for ratios	$\mathbb{Q}$
5. Limits of enclosure sequences	Countable sequences of $\#$ and $[]$	$\mathbb{R}_{\text{comp}}$ (computable reals)
6. Monna-map projection	Project Bruhat-Tits tree onto smooth manifold	$\mathbb{R}$ (continuous shadow)

Steps 1–5 are constructive: every new element is finitely definable from the primitives using finitely many operations. Step 6 — the Monna-map projection — maps the discrete, non-Archimedean Bruhat-Tits tree onto a continuous, Archimedean manifold. This projection is **lossy**: it creates elements — the non-computable reals $\mathbb{R} \setminus \mathbb{R}_{\text{comp}}$ — that have no discrete counterpart in the tree. They are projection artifacts.

4.2 The Stratigraphy of Measurement

The history of number systems is a history of expanding the frontier of what we can distinguish. Each era added a new distinction operation, yielding a richer number system:

Era	Distinction Operation	LoF Primitive	Number System
~30,000 BCE	Mark once, twice, thrice...	Repeated $\#$ (Calling)	$\mathbb{N}$
~500 BCE	Enclose marks, compare ratios	Nested $[]$	$\mathbb{Q}$

Era	Distinction Operation	LoF Primitive	Number System
~1670 CE	Infinite converging sequences	Countable sequences of # and []	$\mathbb{R}_{\text{comp}}$
~1870 CE	Project tree onto smooth manifold	Monna-map (lossy)	$\mathbb{R}$
~1800 CE	Distinguish phase	Imaginary enclosure	$\mathbb{C}$
~1900 CE	Distinguish by divisibility	p-adic enclosure	$\mathbb{Q}_p$
~1950 CE (at once)	All valuations simultaneously	Adèlic enclosure	$\mathbb{A}$

This stratigraphy reveals a crucial asymmetry: the step from $\mathbb{Q}$ to $\mathbb{R}_{\text{comp}}$ (step 5) is constructive, while the step from $\mathbb{R}_{\text{comp}}$ to $\mathbb{R}$ (step 6) is projective and lossy. The non-constructive reals — the vast uncountable majority of $\mathbb{R}$ — are artifacts of the projection, not products of distinction operations. Physics has never needed them.

5. The Gisin–Del Santo Programme: Independent Convergence

5.1 Real Numbers as Hidden Variables

Nicolas Gisin’s core argument (Gisin 2019) [certainty: 5/5 — the argument is a structural observation, not an empirical prediction] runs as follows. Consider a chaotic classical dynamical system. The equations of motion are deterministic, and the entire trajectory $(\vec{x}(t), \vec{p}(t))$ is fully determined by the initial conditions $(\vec{x}(0), \vec{p}(0))$. For chaotic systems, the leading digits of $\vec{x}(t)$ depend on digits far down the series of $\vec{x}(0)$. These far-down digits are inaccessible — no measurement can determine them. Yet standard classical mechanics, by modeling initial conditions as real numbers, asserts they exist as fully determined physical quantities.

Gisin observes that this is structurally identical to hidden-variable theories: the real numbers encode all future states in their infinite digit sequences, exactly as hidden variables would. Since hidden variables are physically unreal, real numbers are physically unreal. Classical mechanics, properly understood, is indeterministic.

5.2 Finite Information Quantities

Del Santo and Gisin (Del Santo and Gisin 2019) develop this into a full alternative theory of classical mechanics. They propose **Finite Information Quantities (FIQs)**: at each point in time, a physical quantity is determined only up to finite precision. The digits beyond that precision are **genuinely indeterminate** — not merely unknown, but ontologically undefined. When a chaotic system amplifies an undetermined digit into a macroscopically relevant one, that digit must become determinate through a process they identify as a classical measurement problem, structurally parallel to the quantum measurement problem.

The alternative theory makes precisely the same empirical predictions as standard classical mechanics. The difference is ontological: where standard theory treats all digits as determined *ab initio*,

the FIQ theory treats undetermined digits as becoming determinate through time-developing processes. This distinction between **geometric time** (deterministic parametrization) and **creative time** (novel information creation) is developed further in subsequent work.

5.3 Ontological Closure Alignment

This is precisely the Autaxys Ontological Closure (OC) boundary between measurable and imaginable. The real number $\sqrt{2}$ is mathematically well-defined (imaginable) but cannot be “measured” — only approximated by a finite D/R procedure (measurable). Gisin’s “real numbers are hidden variables” identifies the same ontological distinction as OC’s “a quantity is physically real only if there exists a finite Turing-machine protocol that approximates it.”

Convergent evidence (independent of QNFO): The Bekenstein bound independently constrains a finite region of spacetime to contain finite information. If real numbers require infinite information to specify, and spacetime regions have finite information capacity, then real numbers cannot be physically realized in any finite region. This is a constraint from general relativity and thermodynamics, not from the adelic programme.

6. Bruhat-Tits Trees as Physical Geometry

6.1 Discrete Substrate of Spacetime

The Bruhat-Tits tree T_p for a prime p is an infinite $(p+1)$ -regular tree that is the natural geometric object associated with the p -adic numbers $\mathbb{Q}_p$. In the adelic framework, the physical geometry at each p -adic place is not a smooth manifold but a discrete tree. Spacetime is not a single manifold — it is a restricted product of geometries, one at each completion of $\mathbb{Q}$.

The product tree $T_2 \times T_3 \times T_5$, with its diagonal embedding into the Pythagorean semigroup $\{2^a \cdot 3^b \cdot 5^c\}$, maps tree geometry to physical mass ratios (Quni-Gudzinis 2026b). The Adelic Cross-Domain Program reports that 11 Standard Model mass ratios fit this scheme to approximately 2%, with specific examples including: - $m_\tau / m_c = 136.63$ vs. $3^7 / 2^4 = 136.7$ (deviation 0.07%) - $m_\tau / m_d = 20.02$ vs. $2^2 \cdot 5 = 20$ (exact)

Convergent evidence (independent of QNFO): Chen, Liu, and Hung (2024) (Chen et al. 2024) independently treat the Bruhat-Tits tree as a physical geometry, constructing a p -adic BTZ black hole on T_p . This provides convergent evidence from a separate research programme that BT trees can function as spacetime geometries.

6.2 Zitterbewegung as $\mathbb{R}/\mathbb{Q}_2$ Topological Mixing

Zitterbewegung (ZBW) is the rapid oscillatory motion of a relativistic electron predicted by the Dirac equation, with frequency $\omega_Z = 2E_p/\hbar \approx 1.6 \times 10^{21}$ rad/s and spatial amplitude on the order of the Compton wavelength $\sim 10^{-13}$ m. The oscillation arises from interference between positive-energy and negative-energy branches of the Dirac Hamiltonian (Gerritsma et al. 2010).

In the adelic framework (Quni-Gudzinis 2026i), ZBW is interpreted as mixing between the ∞ -place and the 2-place. A localized wave packet at x_∞ necessarily contains Fourier components delocalized at x_2 . The Compton-scale oscillation is the physical signature of the $\infty \leftrightarrow 2$ channel. The prime 2 is selected structurally: T_2 is $(2+1)=3$ -regular — the smallest possible Bruhat-Tits

tree — and the Compton scale is the smallest Archimedean scale at which p-adic mixing becomes visible [untested — theoretical prediction requiring experimental verification].

7. Ultrametric Quantum Computation

7.1 Bruhat-Tits Qudits

The QNFO ultrametric quantum computation framework (Quni-Gudzinis 2026h) proposes that the Bruhat-Tits tree can serve as a computational substrate: - Physical qudits (quantum systems of dimension p) reside at vertices of the tree - Hecke operators T_p act as unitary gates moving quantum information along edges - **Passive geometric error correction** from the ultrametric: the non-Archimedean property $\|x + y\|_p \leq \max(\|x\|_p, \|y\|_p)$ prevents small errors from accumulating

The p-adic quantum Fourier transform is realized as a sequence of radix- p quantum gates on Bruhat-Tits qudits. For $p = 2$, this is the Walsh-Hadamard transform requiring no complex multiplications.

7.2 The Adelic FFT Processor

An adelic FFT processor — a physical device that performs harmonic analysis on the adèle ring — would operate as follows:

1. **Local preparation:** Encode the input function f_p in qudits on the Bruhat-Tits tree for each prime $p \leq N$
2. **Local transform:** Apply p-adic quantum Fourier transform gates
3. **Hecke coupling:** Apply Hecke operators to enforce the global restricted-product constraint — this step couples all places simultaneously and is the computational bottleneck
4. **Archimedean readout:** Physical measurement at the Archimedean place, producing f_∞ as the observable output

The key architectural insight is a hardware/software co-design: p-adic computations are noise-protected by the ultrametric (passive QEC), while the Archimedean readout interfaces with human observers. The adelic nuclear-spin qubit architecture operates at room temperature (Quni-Gudzinis 2026f), eliminating the multi-million-dollar dilution refrigerator requirement that constrains superconducting qubit platforms [speculative — hardware specification, not yet experimentally validated].

8. The Langlands Bridge

8.1 Langlands Physics via S-Duality

The Langlands Program — a grand web of conjectures linking number theory and geometry — connects with quantum field theory and string theory through electric-magnetic S-duality (Kapustin and Witten 2006):

- In $\mathcal{N} = 4$ super Yang-Mills theory, S-duality swaps electric and magnetic fields and interchanges the gauge group with its Langlands dual group
- Kapustin and Witten demonstrated that this quantum physical symmetry matches the mathematical framework of the geometric Langlands correspondence (Kapustin and Witten 2006)

- Physical concepts (quantum branes, Wilson operators, 't Hooft operators) map directly onto esoteric geometric objects (Hecke eigensheaves, D-modules)

Physical significance: The Langlands correspondence provides a dictionary between physics and number theory at the deepest structural level. The adelic programme provides a concrete physical interpretation of this dictionary: the Langlands correspondence is the statement of S-duality on the adèle group $G(\mathbb{A})/G(\mathbb{Q})$, where automorphic representations form the physical Hilbert space and Galois representations encode the symmetry algebra (Quni-Gudzinis 2026a).

8.2 FFT as Computational Langlands

The Fast Fourier Transform (FFT) on the additive group $\mathbb{R}/\mathbb{Z}$ has a precise interpretation as the Langlands correspondence for $GL(1)$ at the Archimedean place (QNFO Research 2026). The adelic FFT processor extends this interpretation: the global FFT on the adèle ring is the computational realization of the Langlands correspondence for $GL(1)$ at all places simultaneously. This provides a concrete, computable entry point into the Langlands programme for physicists — the FFT they already know is Langlands in disguise.

9. Domain Translation Errors: A Communications Framework

9.1 The Vocabulary Problem

Quantum computing borrows its vocabulary from at least four distinct physical domains — quantum optics, condensed matter physics, the Standard Model of particle physics, and classical computing — and the terms do not translate cleanly. The “photon” of circuit QED is not the photon of quantum optics; the “electron” in a spin qubit is not the electron of the Standard Model. These category errors compound into a systematic map-territory confusion [uncontested observation — established by working practitioners].

The pedagogical-versus-operational distinction is critical: working quantum engineers do not believe qubits are tiny billiard balls. They think in terms of collective excitations of modes, relational observables, and spectral addressability. The confusion is in textbooks, press releases, and funding narratives — not in the design of quantum processors.

9.2 The Adelic Programme’s Communication Challenge

The adelic programme faces a more severe version of this problem. Its vocabulary is borrowed from number theory (valuations, completions, adeles, Hecke operators) — a domain that most physicists have never studied. The challenge is not that the programme is wrong; it is that its vocabulary is foreign.

This paper is part of the solution: by grounding the argument in concepts physicists already know (computable reals, finite-precision measurement, the Bekenstein bound, the FFT), we make the adelic programme accessible without requiring a detour through algebraic number theory. The fundamental claim — “ $\mathbb{Q}$, not $\mathbb{R}$, is the physical base field” — requires no adelic vocabulary to state.

9.3 Metrological Independence

A related communications issue: quantum advantage claims should specify how calibration circularity has been bounded, what independent anchors have been used, and what the residual calibration uncertainty is [proposed methodological standard — untested in practice]. This connects to the broader theme of measurement limits in the $\mathbb{Q}$ framework: if all physical quantities are $\mathbb{Q}$ -approximable, then every measurement protocol must explicitly bound its calibration circularity.

10. Observer-Centered Epistemology

10.1 The Zero Point of Observation

In every act of measurement, representation, or expression, there is an assumed zero point — the observer’s “here, now, and thus.” It is so natural we forget it is a choice. The choice of $\mathbb{R}$ as the base field of physics is precisely such a zero point: the Archimedean place is the “here” from which we view the mathematical landscape, and we mistake this perspective for the landscape itself.

Incompleteness is the shadow cast by a single observer mistaking their perspective for the whole. Completion is the discovery of the transformation rules that relate all perspectives, and the invariant core that remains. The adelic programme is that discovery: the adèle ring $\mathbb{A}$ is the space of all perspectives (all completions of $\mathbb{Q}$), and the restricted product is the transformation rule that relates them.

10.2 Epistemic Humility

The deepest hidden assumption is that our certainty has cosmological reach. We habitually take a locally validated model — base-10 arithmetic, classical mechanics at human scales, quantum field theory at accelerator scales — and extend it indefinitely. The mistake is not in the model; it is presuming that the conditions under which it was validated hold everywhere and always. This is epistemic ignorance at the most fundamental level.

The adelic programme is an exercise in epistemic humility: it asks what happens when we stop presuming that the Archimedean completion is the unique “right” way to complete $\mathbb{Q}$, and instead treat all completions as equally legitimate. The result is not a rejection of standard physics — the Archimedean place remains central to measurement — but an expansion of the physical ontology to include what was always there, hidden by an unjustified assumption.

11. Conclusion: What Changes

If $\mathbb{Q}$ is the physical base field:

1. **The Adelic Physics Program transitions from “mathematical speculation” to “physical programme.”** Ostrowski’s theorem becomes a physical organizing principle, and the p-adic completions are physically meaningful.
2. **The Ostrowski Programme clears.** The deepest unexamined premise of physics — that $\mathbb{R}$ is the natural base field — is examined and defended. The adelic programme is the consequence of taking that examination seriously.

3. **Hensel Codes are not merely a computational trick.** They reflect the actual $\mathbb{Q}$ -based structure of arithmetic. p -adic numbers are not exotic — they are completions of the same $\mathbb{Q}$ that physicists use every day.
4. **The Gisin–Del Santo programme converges with OC.** The argument that real numbers are hidden variables and that physics requires only finite information quantities provides independent, non-QNFO support for the $\mathbb{Q}$ -as-base-field thesis.
5. **QNFO physics can advance without the base-field liability.** The adelic programme is not “new physics” — it is the removal of an unjustified assumption that has constrained physics for 150 years.

The burden of proof has shifted. The advocate of $\mathbb{R}$ must now demonstrate a physical prediction that requires infinite-precision real numbers and cannot be reproduced by any finite-precision rational computation. Until such a prediction is found — and none has been found to date — the working assumption should be that $\mathbb{Q}$ is the physical base field, and that Ostrowski’s theorem applies.

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